PREFACE

Much of what has been written on the topic of the National Aeronautics and Space Administration's (NASA) tracking and data networks has been on the Jet Propulsion Laboratory's Deep Space Network, the DSN. This is perhaps understandable as the DSN has played and continues to play a central role in many of America's most high-profile exploration missions. These have included the early Pioneer probes, the Mariner missions of the 1960s and 1970s, Viking and Voyager, and most recently, Galileo, Cassini-Huygens, and the new generation of Mars explorers that will prepare the way for eventual human voyages to the Red Planet.

The intent of this volume is to present a history of NASA's "other" network, the one established and run by the Goddard Space Flight Center (GSFC). The Spaceflight Tracking and Data Network, or STDN, was—in its various incarnations throughout the years—the network that tracked the first artificial satellites around Earth. It tracked Apollo astronauts to the Moon and back. Today, a network based in space called the Space Network, along with a much reduced Ground Network, work together to support the United States and international partners in all near-Earth space communications and space-flight activities. The history of the STDN is not unlike a microcosm of the history of NASA itself. It spans 50 years. It has seen its share of triumphs and tragedies, and it is playing a major role in setting the pace for space exploration in the twenty-first century.

When considering sources for this history, the author searched for scholarly works that have been published on the subject of NASA's STDN.

There has been some coordinated effort to document NASA's human spaceflight and near-Earth communications networks throughout the years. Starting in the late 1960s, the GSFC saw the need to begin documenting this history. From that start, most of the literature on the subject has been independently commissioned reports sponsored by the Center itself.

The seminal work was by William R. Corliss in 1969 called *History of the Goddard Networks*. Corliss updated it five years later, expanding the subject to the end of the Apollo program. These initial works were relatively general, based primarily on secondary research—assimilation of information put out by the GSFC in the form of information pamphlets, brochures and public affairs news releases.

NASA historian Alfred Rosenthal drafted, in 1983, an unpublished work titled *Vital Links: The First 25 Years of NASA's Space Tracking, Communications and Data Acquisition 1958–1983. Vital Links* was a new work on the subject, taking the timeline up through the early part of the Space Shuttle program. A key part of Rosenthal's work was his interview of some 20 people whom he identified as principals on the subject. These were not formal oral histories but rather, topic-specific statements made by people who were major contributors on the STDN over the years. The interviews were done in 1982 and 1983 but never published. Many of these people are now deceased. The author drew heavily on these interviews and many of the quotes are used throughout this book. The draft of *Vital Links* was turned over to the NASA History Division where it is archived. It was a primary reference for this work.

In 1992, GSFC published the Contractor Report *Keeping Track: A History of the GSFC Tracking and Data Acquisition Networks 1957 to 1991*. It was edited by Kathleen M. Mogan and Frank P. Mintz. The report listed them as editors rather than coauthors because *Keeping Track* was, for the most part, edited together from Corliss's 1974 and Rosenthal's 1983 works. A final chapter was added which provided a very brief overview of the STDN in the 1980s. Mogan and Mintz included excerpts from about half of Rosenthal's interviews at the end of the report in a section called "Personal Views." They, in addition, conducted half a dozen interviews with key network personnel who were at Goddard in 1990. The material was then published by the GSFC as a reference report where it has circulated since. Copies have also been distributed to other NASA locations. Therefore, through Rosenthal, Mogan, and Mintz, over two dozen people who shaped NASA's STDN from 1960 to 1990 were interviewed. This was an invaluable resource. Taken together, *Vital Links* and *Keeping Track* served as the primary reference, the updating of which was an impetus for this book.

The most important archival materials were undoubtedly the Historical Reference Collection at the History Division of NASA Headquarters. There,

correspondences and memoranda on the subject were found dating back to 1958. Letters between key Administrators and managers like Edmond C. Buckley, Gerald M. Truszynski, and Ozro M. Covington—early movers of the program—explained why certain decisions were made and how they arrived at those decisions. Files were found all the way to the present. Organized by subject and key words, files addressing a specific topic with regard to the Networks were found in the form of Congressional testimonies, news articles, technical presentations and photographs. These included those published by the Agency over the years as well as those from open literature such as newspapers, magazines and professional journal articles.

Technical briefings and presentations provided to the author by NASA was another good source of firsthand information. Most of these came from the Office of Space Communications in Washington and some from the GSFC. The author also interviewed current and former NASA managers and others who worked with the Network, both inside the government and contractors, both U.S. and foreign. They brought unique, personal perspectives that answered questions or helped clarify ambiguities uncovered in the course of the research. These personal perspectives also provided stories which simply could not have been found anywhere else. NASA Oral Histories—most of which are now available online—provided much insight to this end. They complimented, and in certain cases, supplemented the histories obtained from other sources. Most of these Oral Histories were conducted by the Agency as part of the ongoing Johnson Space Center (JSC) Oral History Project started in 1998. Interviews from three groups of people were reviewed: NASA Administrators and high level managers, flight controllers and operations personnel, and astronauts. This produced a broad and diverse set of viewpoints from which a story could be weaved together, supplementing the material uncovered from other sources.

Books on the subject of tracking networks, space communications, space exploration, and political history in general provided good background information within which the topic was framed. Even though these generally lacked the details necessary to delve in-depth into the actual history of Networks, they provided the general science and historical background which complemented this history, and which in fact, allowed the story to be told from a popular point of reference. Other books of a more technical nature provided the material needed to explain some of the finer scientific points in layman terms.

Transcripts such as Administrator testimonies before Senate and House subcommittees and White House letters were examined. The World Wide Web provided convenient material, but they required verification from other sources. Most could be validated by authenticating or tracing back to the original Web site. Those from dubious sources were either not used at all or

if used, clarified in more detail in the Endnotes section of this book. Internet sites were especially helpful for general historical information on topics such as history of a country, the climate of an island, or the evolution of a country's governing body and its political relationship to the United States.

A diverse range of photographs and pictures are included in this book. When discussing the topic of tracking stations and communication networks, it is easy to fall prey to reproducing pictures of big antennas and buildings page after page. After all, those are probably the most prominent components of a ground station. Some of those are included. But what paints a more interesting and complete picture are the people involved and the environment that they operated in. Therefore, a deliberately chosen mix of seldom before seen photographs along with better-known, more frequently reproduced pictures are included.

The history of NASA's STDN is a 50 year story, from its birth in the late 1940s to where it stands today in the first decade of the twenty-first century. The story is told here from a historical and not a technical point of view, although explanatory passages are present when warranted. The author hopes that in doing so, the chapters will appeal not only to the trained subject expert but also to a lay audience or just the space enthusiast at large who may simply have always been curious as to how NASA tracked the Lunar Rover on the Moon, for instance.

Since the beginnings of the STDN are inextricably tied to the formation of NASA as America's civilian space agency, a brief synopsis of how the Agency came about—the nuances of which affected how it would operate its tracking networks for years to come—is included in Chapter 1. For a comprehensive look at the establishment of NASA, the author recommends starting with *NASA's Origins and the Dawn of the Space Age*, Monographs in Aerospace History number 10, by David S. F. Portree, 1998.

Similarly, the history of the STDN is tied to the Space Shuttle. After the Apollo lunar missions, the Network turned its focus on supporting the reusable Shuttle. Much has been written on the history of the Space Transportation System (STS), so much so that in 1992, NASA's History Division published a monograph summarizing the vast amount of literature on the subject. The reader is encouraged to reference that work for more information. (Launius, Roger D. and Gillette, Aaron K., *The Space Shuttle: An Annotated Bibliography*, Monographs in Aerospace History, Number 1, 1992)

Finally, this book does not address the Deep Space Network DSN except where it overlaps and affects the STDN. Works on the Deep Space Network include *A History of the Deep Space Network* by William R. Corliss from 1976, and the 2001 NASA Project Histories publication SP-2001-4227, *Uplink-Downlink: A History of the Deep Space Network 1957-1997* by Douglas J. Mudgway. The latter work is a very comprehensive description of the DSN from a technical perspective. Mudgway followed this volume with a less tech-

nical but more historical look in his 2005 publication *Big Dish*. Since it is difficult to fully segregate the two networks, the following is an overview of the DSN which the reader may find beneficial as he or she delves into the history of the STDN in the coming chapters. Much of this description can be found in the 1989 Jet Propulsion Laboratory publication JPL400-326 *Goldstone Deep Space Communication Complex:*

Scientific investigation of the planets and interplanetary space has been carried out for over four decades by NASA mainly through the use of automated space probes and robotic vehicles. Although engineered to operate independently in the far reaches of space, these intricate and highly autonomous craft are dependent upon Earth-based DSN for guidance, control and reception of the vast amounts of scientific information they acquire and transmit back to Earth.

NASA's DSN is today among the largest and most sensitive scientific telecommunications and radio navigation networks in the world. Its principal responsibilities are to support unmanned planetary missions and radio and radar astronomy in the exploration of the solar system and the universe. The DSN is managed, directed and operated for NASA by the Jet Propulsion Laboratory (JPL) of the California Institute of Technology in Pasadena, California. It performs several, wide-reaching functions, all relating to the deep space environment: 1) Receive telemetry signals from space probes; 2) Transmit commands that control the spacecraft; 3) Generate radio navigation data to locate and guide the spacecraft to their destinations; 4) Perform radio science, radio and radar astronomy; 5) Perform highly sensitive Earth geodynamics measurements; and 6) Participate in the Search for Extraterrestrial Intelligence.

The history of the DSN dates back to 1958 when JPL deployed three ground communication stations to help receive telemetry and determine the orbit of Explorer 1, the first successful American satellite. Before that, engineering research into the tracking requirements for lunar and planetary spaceflight had been in progress for some time at JPL. Following Explorer 1, JPL developed a ground communications and navigation network for the Pioneer 3 and 4 missions, the first U.S. spacecraft to be launched beyond Earth orbit. On 3 December 1958, shortly after the establishment of NASA, JPL was officially transferred to the new space agency from the U.S. Army. Since then, it has grown with the U.S. space program into a world network consisting of 12 deep-space stations located at three communication complexes on three continents: Goldstone in Southern California's Mojave Desert; at Robledo near Madrid, Spain; and at Tidbinbilla on the outskirts of Canberra, Australia. These three locations allow the DSN to maintain around-the-clock communications with spacecraft traveling anywhere in the solar system.

The network consists of several key components. One is the Network Operations Control Center (NOCC), which controls and monitors

the three complexes. Located in Pasadena, the NOCC is the operations hub for the DSN. Its functions include monitoring operations at the three complexes, validating the performance of the network for spaceflight project users, providing information for configuring and controlling the network, and planning and participating in network testing and mission simulations.

The Ground Communications Facility provides the necessary communication circuits that link the complexes to the control center at JPL and to the remote flight project centers in the United States and overseas that manage and operate the space probes. "Comm" traffic between these places can be sent in various ways, such as via land lines, ocean cables, terrestrial microwave, and communication satellites. These circuits are leased from commercial carriers by the NASA Integrated Support Network, commonly referred to as NISN—jointly managed by the Goddard and Marshall Space Flight Centers—which provides circuits as needed to all NASA centers and facilities.

The most visible arm of the DSN is of course the three Deep Space Communication Complexes that are dominated by a suite of large parabolic dish antennas. At each complex are two 34-meter (111-foot) diameter antennas, one 26-meter (85-foot) antenna, and one gigantic 70-meter (230-foot) antenna. A centralized Signal Processing Center remotely controls the 70-and 34-meter antennas, generates and transmits spacecraft commands, and receives and processes the spacecraft telemetry. These antennas form separate subnets according to their respective sizes, and each subnet has a different communication capability. For example, the 70-meter antenna subnet—which is the most sensitive—supports deep space missions while the 26-meter subnet supports spacecraft in near-Earth orbit. The two 34-meter antenna subnets support both types of missions.

The 26-meter systems were originally part of the Goddard STDN. They were consolidated into the DSN in 1985. The merger led to significant operational efficiencies by consolidating like maintenance, operations, and field support activities for tracking stations located geographically close to each other. From a mission's perspective, it also allowed the DSN to now track spacecraft in highly elliptical, high apogee Earth orbits.

In the United States, the DSN is staffed by JPL personnel, assisted by contractor engineers and technicians who are primarily responsible for operating and maintaining the Goldstone complex, the NOCC and the Ground Communications Facility. The two overseas complexes are operated entirely by the local government agency (the National Institute of Aerospace Technology, or INTA, in Spain and the Commonwealth Scientific and Industrial Research Organization, or CSIRO, in Australia) and their contractors under NASA funding. The international staff totals over 1,000 people.

Since its beginning, the DSN has provided principal tracking, telemetry, and command (TT&C) support for every unmanned lunar and planetary spacecraft that NASA has launched (plus secondary support on Apollo). The resumé is quite impressive. Moon exploration began with the Pioneer 3 and 4 probes in 1958 and 1959. This was followed in the 1960s by the Ranger 1 through 9 lunar television missions (1961–1965); the Lunar Orbiter 1 through 5 photographic surveys for Apollo landing sites (1966–1967); and Surveyor 1 through 7 lunar landers (1966–1968). Support for inner-planet exploration began in 1962 with JPL's Mariner series of missions to Venus, Mars and Mercury that encompassed six planetary flybys and the first planetary orbiter around Mars in 1971. Concurrent with these Mariner flights were the NASA Ames Research Center Pioneer missions, Pioneer 6 through 9 between 1965 and 1968. Mars exploration culminated in 1975 with the much-heralded landings of Viking 1 and 2.

The first JPL probe to explore the outer planets was Pioneer 10, which flew by Jupiter in 1973. Six years later, Pioneer 11 became the first spacecraft to flyby Saturn. Two Voyager probes performed a grand tour of the outer planets following their launch in 1977, passing Jupiter in 1979 and Saturn in 1980 and 1981. Voyager 2 then went on to make close approaches to the seventh planet, Uranus, in 1986 and three years later Neptune. In both cases, stunning close-up photographs of these distant planets were beamed back to Earth. They are now far beyond Pluto's orbit and astronomers anticipate they will venture beyond our known solar system in the next few years.

More recently, the DSN is supporting the Cassini-Huygens mission, exploring Saturn and its largest moon Titan. The latter is one of the most intriguing objects in the entire solar system. The second largest of any of the planetary moons (only Jupiter's Ganymede is larger), Titan is actually larger than Mercury and Pluto. It, in fact, has a planet-like atmosphere which is denser than those of Mercury, Earth, Mars, and Pluto. Composed predominantly of nitrogen with other hydrocarbon compounds, such elements are the essential building blocks for amino acids necessary for the formation of life. This has led some scientists to postulate that Titan's environment may be similar to that of Earth before life began putting oxygen into the atmosphere.

Finally, the DSN is once again supporting new NASA missions to our nearest planetary neighbor, this time with the Mars Reconnaissance Orbiter along with the Spirit and Opportunity robotic rovers. This is a prelude to eventual human exploration of the planet that is slated for the 2030 to 2040 timeframe. Pluto remains the only planet (dwarf planet) not having been visited by a manmade spacecraft. But that will change by the year 2015

as the New Horizons spacecraft—launched on 19 January 2006—is scheduled to encounter our solar system's most distant planet. It will then make an extensive study of as many Kuiper Belt objects as it can, a mysterious disk shaped zone beyond Neptune containing thousands of planetary-like objects, before permanently leaving the confines of the solar system to travel into interstellar space.[1]